- **AXLE—REAR AXLE "WHINE" NOISE—VEHICLES EQUIPPED WITH DANA 80 AXLES—5.4L/6.8L VEHICLES BUILT THROUGH 5/24/2000 AND 7.3L VEHICLES BUILT THROUGH 12/18/1999**
- **NOISE—REAR AXLE "WHINE"—VEHICLES EQUIPPED WITH DANA 80 AXLES—5.4L/6.8L VEHICLES BUILT THROUGH 5/24/2000 AND 7.3L VEHICLES BUILT THROUGH 12/18/1999**

Article No. 00-15-4

FORD: 1996-2000 ECONOLINE

ISSUE

Some E-450 Super Duty Cutaway/Stripped Chassis vehicles with Dana 80 rear axles may exhibit an excessive amount of axle "whine" noise. Replacement of the axle is not always necessary. Axle whine often is corrected by replacing only the rear driveshaft of the driveshaft assembly with a dampened rear driveshaft. The dampened rear driveshaft should always be installed before the axle is replaced.

ACTION

Install a dampened rear driveshaft and evaluate for the amount of improvement. If improvement does not meet customer expectation, then replace the axle assembly. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

NOTE

DANA MODEL 80 AXLES WERE REPLACED IN PRODUCTION BY DANA MODEL 70 T-RIB AXLES ON:

ENGINE/RATIO	CHANGE-OVER DATE
7.3L/4.10 5.4L/6.8L/4.63	December 18, 1999 May 22 and 23, 2000 (4.63, Mixed built on the 22nd and 23rd)

Hints for determining the type of axle: Dana 80 axles were only used on E-Series units where the 5th position of the VIN is "E" and the 6th position of the VIN is "4" (E40, E45, E47, E49). The parking brake position also provides a way to ensure that you are working on an E-450. All E-450s have the parking brake at the back of the transmission on the driveshaft. If there is not a parking brake assembly on the driveshaft, the vehicle being worked on is not an E-450 and this TSB does not apply.

To determine if the axle (2000 model year E-450 only) is a Dana 80 (Figure 1) or Dana T-Rib 70 (Figure 2), look at the pinion flange.

CAUTION

DO NOT REDUCE PINION BEARING PRELOAD IN AN ATTEMPT TO REDUCE AXLE WHINE. THIS HAS BEEN A COMMON (UNAUTHORIZED) FIELD PRACTICE. REDUCING PINION BEARING PRELOAD MAY RESULT IN REDUCED AXLE DURABILITY.

NOTE

THE DAMPENED DRIVESHAFT IN THIS TSB IS DESIGNED TO REDUCE THE SOUND LEVEL OF AXLE WHINE CONCERNS. THE DAMPENED DRIVE IS NOT INTENDED FOR OR TO BE USED TO ATTEMPT TO RESOLVE A DRIVELINE VIBRATION CONCERN.

1. Verify the customer concern by road testing the vehicle. HINT: The most common speed ranges where customers have experienced E-450 axle whine are 64-80 km/h (40-50 mph) and 96-112 km/h (60-70 mph).
2. Determine the original wheelbase. The original wheelbase may be found in the lower left corner on the Vehicle Certification Label (Ford Incomplete Vehicle Label). E-450s are built with only a 158 inch or 176 inch wheelbase with a two-piece driveshaft. A large number of E-450s are modified/stretched by second stage manufacturers. Most modifiers that extend the wheelbase do not change the length of the rear driveshaft.

NOTE

IF THE LENGTH OF THE REAR SHAFT HAS BEEN MODIFIED, A DAMPENED DRIVESHAFT WILL NOT BE AVAILABLE AND REPLACEMENT

OF THE AXLE WILL BE THE ONLY AVAILABLE REPAIR. WITHOUT THE DAMPENED DRIVESHAFT, SOME AXLE WHINE MAY STILL BE PRESENT.

3. Order and install dampened Driveshaft Kit (rear shaft only) by application of the original wheelbase. Refer to the Driveshaft Application Table at the end of this TSB.
4. Test drive the vehicle to determine the effectiveness of the dampened driveshaft. If sound is reduced, return the vehicle to the customer. If sound is not reduced to a level that is acceptable to the customer, continue with the service procedure.
5. Replace the rear axle assembly (less brake assemblies) if the vehicle was built before January 1, 2000. Units built after January 1, 2000 have the same level axle assembly that is available in service. Follow repair procedures starting in section 205-02A of the E-Series Workshop Manual for axle assembly removal and installation.

NOTE
IF THE VEHICLE IS A 1996 TO 1998 MODEL YEAR VEHICLE, THE AXLE FLANGE WILL NEED TO BE EXCHANGED BETWEEN THE NEW AND OLD AXLES. (THE AXLES MAY ONLY BE BOUGHT WITH THE CURRENT "CIRCULAR FLANGE".) FOLLOW THE INSTRUCTIONS IN SECTION 205-02A (REAR DRIVE AXLE/DIFFERENTIAL - DANA 60, 70 AND 80) WHEN EXCHANGING THE AXLE FLANGE.

CAUTION
ALWAYS INSTALL A NEW WASHER AND PINION/LOCK NUT. ENSURE THAT THE PINION/LOCK NUT IS TORQUED TO 637 N•m (470 LB-FT).

6. Test drive the vehicle to confirm the effectiveness of the repair.

NOTE
A DAMPENED DRIVESHAFT MAY NOT REDUCE AXLE WHINE IF THE DRIVESHAFT ANGLES ARE EXCESSIVE OR THE DRIVESHAFT ASSEMBLY IS NOT BALANCED OR IS OUT OF PHASE. IF EXCESSIVE AXLE WHINE STILL EXISTS AFTER THE DAMPENED DRIVESHAFT AND IMPROVED AXLE ASSEMBLY IS INSTALLED, ENSURE THAT THE DRIVESHAFT ASSEMBLY DOES NOT HAVE EXCESSIVE ANGLES OR IS OUT OF BALANCE OR PHASE. REFER TO SECTION 205-00 (DRIVELINE SYSTEM - GENERAL INFORMATION) OF THE E-SERIES WORKSHOP MANUAL IF ANY OF THESE CONDITIONS ARE SUSPECTED.

Model Year	Build Date	Wheelbase	Service Part Number	Length of Rear Driveshaft - Center of U-joint to U-joint
1996-1998		158 inch	F7UZ-4W602-AA	116.4 cm (45.84 inch)
1999	Job 1 to 12/23/98	158 inch	XC2Z-4W602-AA	115.4 cm (45.46 inch)
1999	From 12/24/98	158 inch	XC2Z-4W602-CA	115.4 cm (45.46 inch)
1996-1998		176 inch	F7UZ-4W602-BA	162.0 cm (63.80 inch)
1999	Job 1 to 12/23/98	176 inch	XC2Z-4W602-BA	161.1 cm (63.42 inch)
1999	From 12/24/98	176 inch	XC2Z-4W602-DA	161.1 cm (63.42 inch)

FIGURES OF DRIVESHAFT FLANGES

Figures of rear driveshaft flanges are included in this TSB to aid in determining the correct dampened driveshaft to order. Driveshafts in 1996-1998 model year vehicles had the "Dog Bone" flange (Figure 3) and the "Straight Sided" slip yoke (Figure 5).

Driveshafts built Job 1, 1999 to 12/23/1998 have the "Circular" rear flange (Figure 4) and the "Straight Sided" slip yoke (Figure 5).

Driveshafts built 12/24/1998 and forward have the "Circular" rear flange (Figure 4) and the "Involute Spline" slip yoke (Figure 6).

NOTE
THE DAMPENED DRIVESHAFT KITS CONTAIN THE REAR DRIVESHAFT, THE SLIP YOKE BOOT, MOUNTING HARDWARE FOR THE SLIP YOKE BOOT AND A 13mm SPACER. THE 13mm SPACER MAY BE USED TO LOWER THE CENTER BEARING TO IMPROVE DRIVESHAFT ANGLES IF NEEDED. THE FRAME CROSSMEMBER THAT THE CENTER BEARING

ATTACHES TO WAS LOWERED BY 13mm ON FEBRUARY 21, 1999. DO NOT USE THE 13mm SPACER ON UNITS BUILT AFTER FEBRUARY 21, 1999.

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 509000, 597997, 702000, 702200, 702300

PART NUMBER	PART NAME
YC2Z-4006-BA	Axle Assembly (6.8L/5.4L Engines With 4.63 Axle Ratio)
YC2Z-4006-AA	Axle Assembly (7.3L Engines With 4.10 Axle Ratio)
E8TZ-4320-A	Pinion/Lock Nut
E8TZ-4329-A	Washer

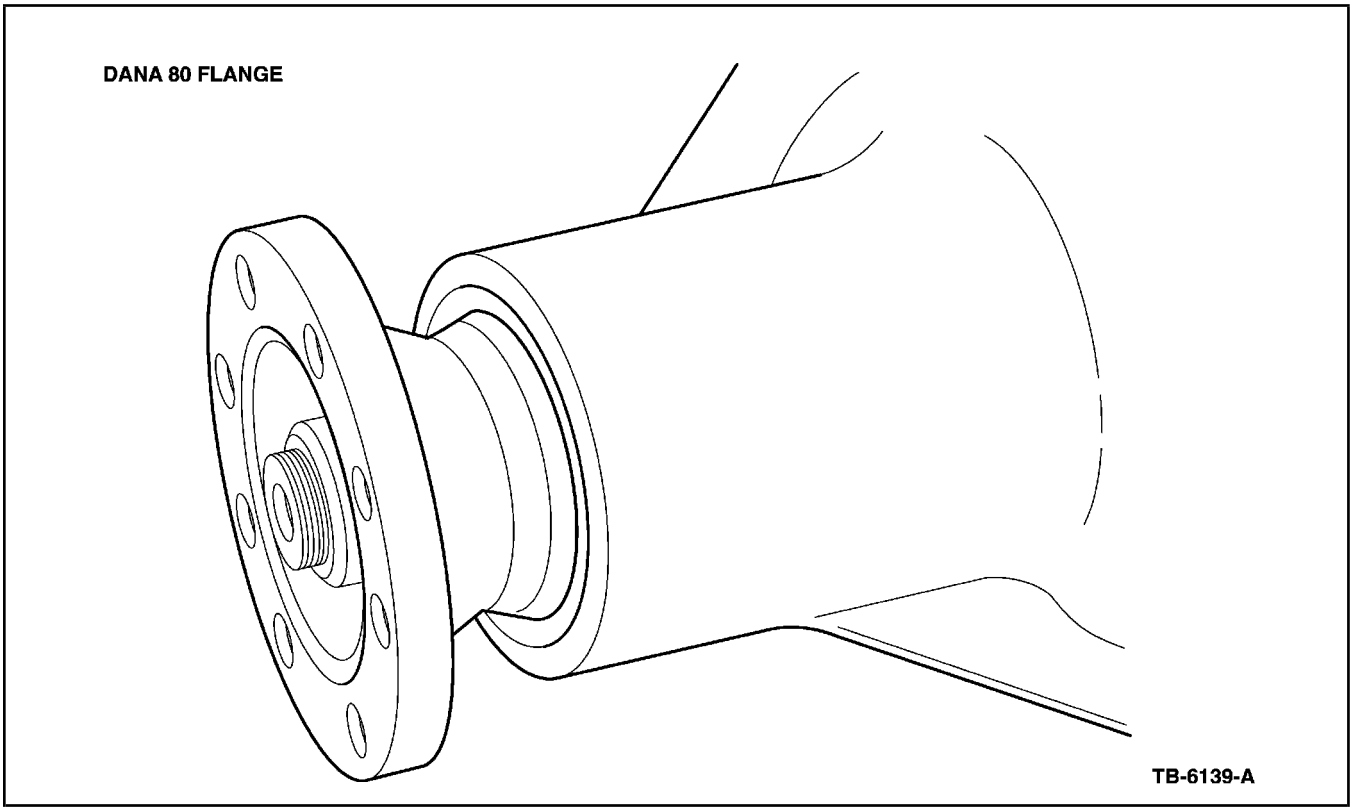

Figure 1 - Article 00-15-4

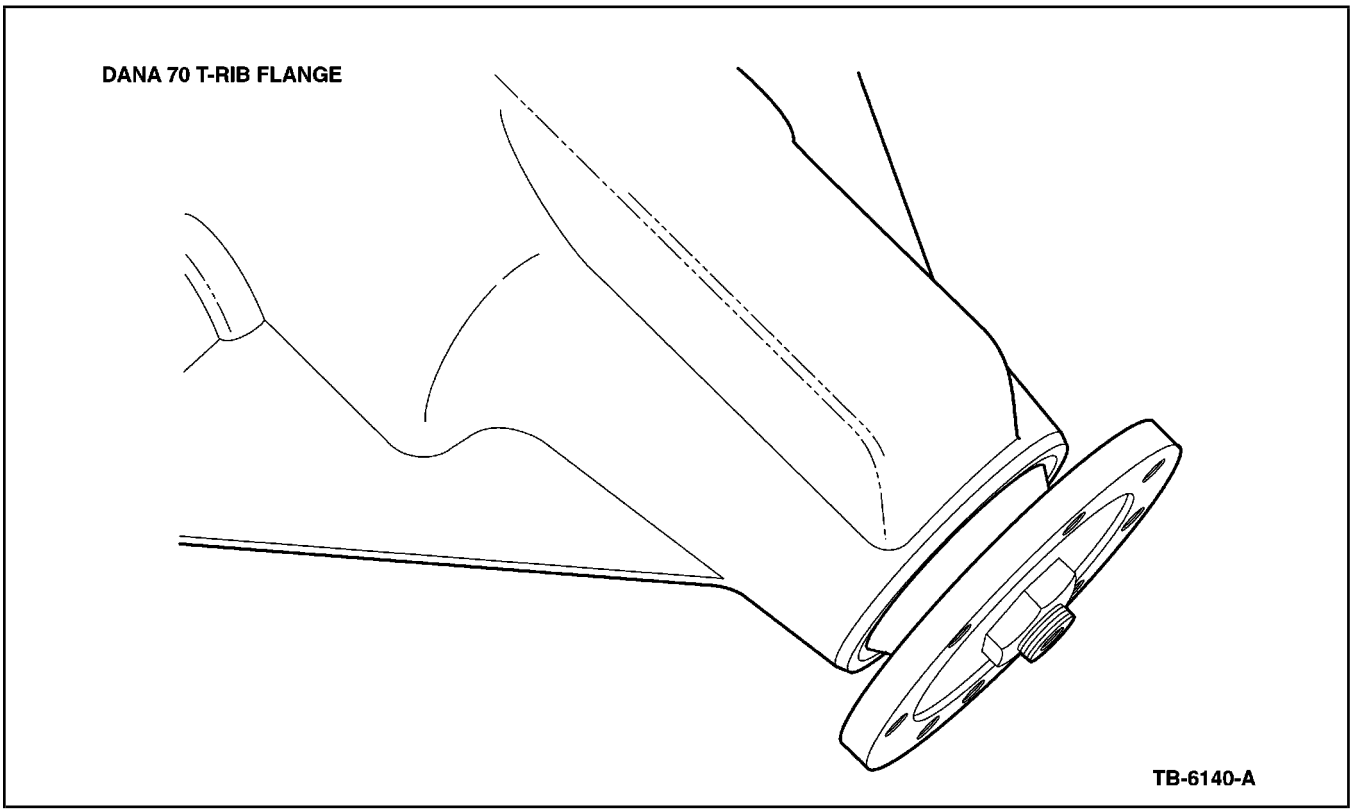

Figure 2 - Article 00-15-4

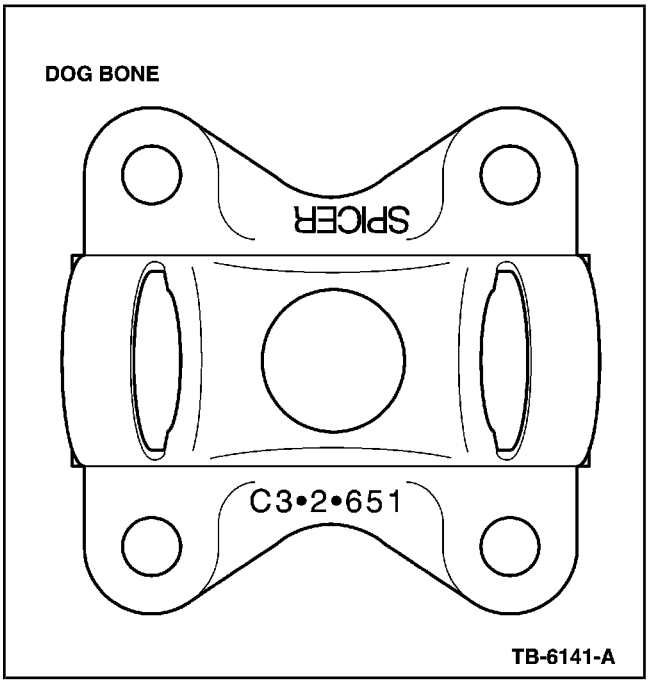

Figure 3 - Article 00-15-4

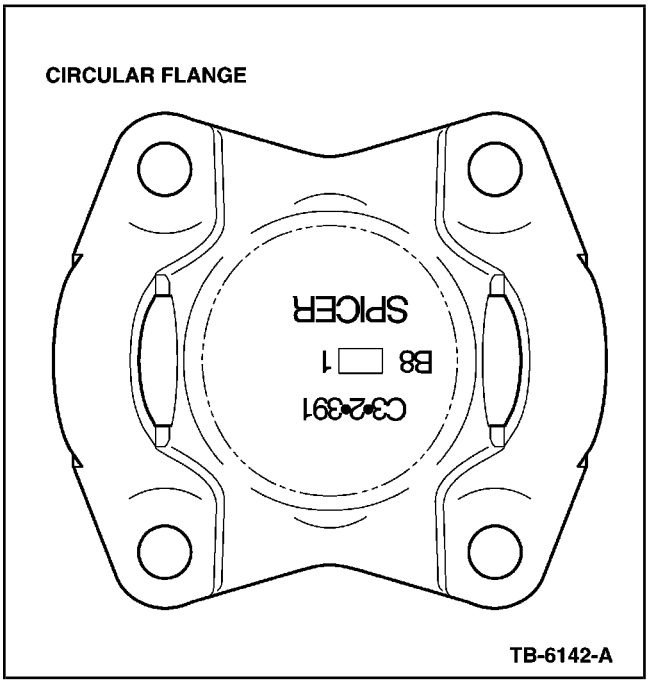

Figure 4 - Article 00-15-4

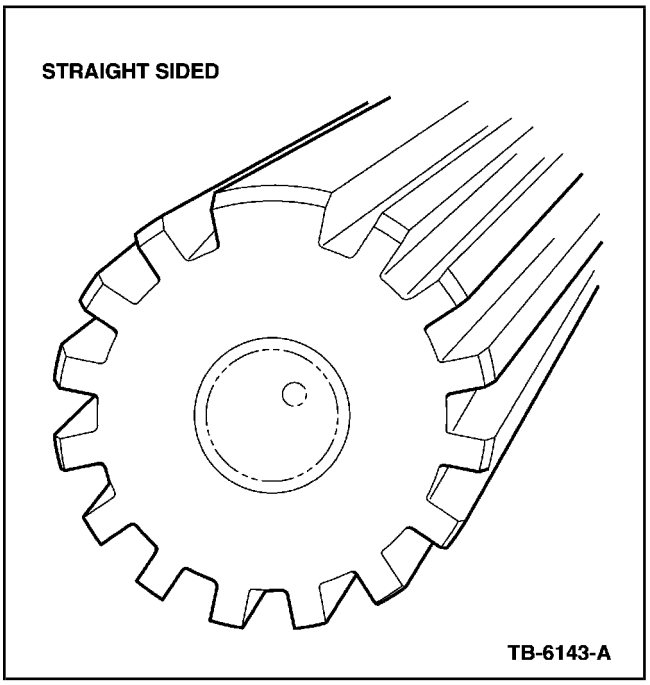

Figure 5 - Article 00-15-4

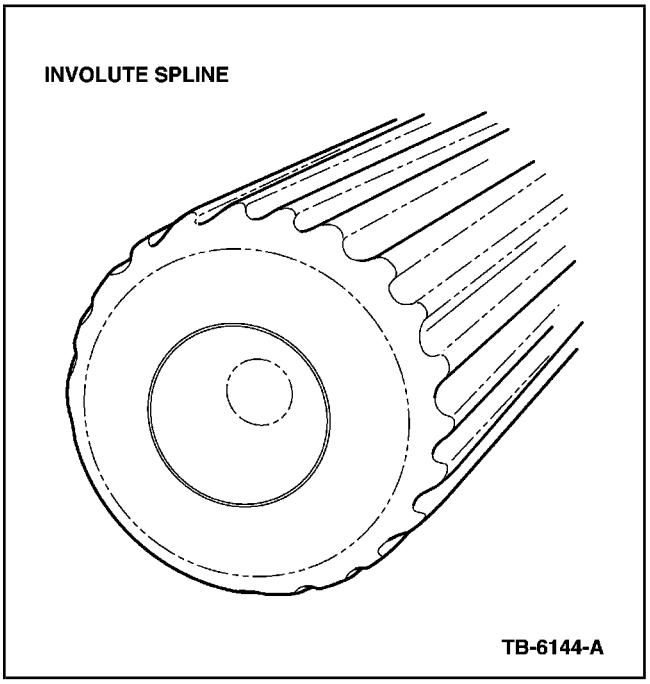

Figure 6 - Article 00-15-4